

Bernard Lappay

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WORK EXPERIENCE

Silent Studios

Jun. 2026 – Present

Technical Contributor, Mineshoku Tensei

Remote

- Cut a core spatial-query system's data complexity by **~86%** by diagnosing the bottleneck and building a Python geometry-simplifier.
- Closed a time-of-check/time-of-use exploit across dozens of config files by scripting a diff-verified patch rollout.
- Shipped race-free real-time boss encounters by driving multi-stage attacks with timing-based state machines and ordered failsafes.
- Moving **40+** world entities from ephemeral spawn/despawn to persistent, state-driven designs, probe-testing each before rollout.
- Audited **2,295** default permissions empirically by building a Python automation harness with checkpointing and pause/resume.
- Keep a multi-node deployment stable across platform updates by diagnosing JVM dependency/version conflicts and logging regressions.

PROJECTS

Adaptive Math-Reasoning Inference Pipeline

Python, vLLM, LoRA, RunPod H100s | CSE 151B, Spr. 2026

- Ranked **#29** class-wide on Kaggle (68/100 final score) by improving a fixed 4B model (Qwen3-4B-Thinking-2507) at math through inference-time design instead of retraining.
- Raised multiple-choice accuracy from **73.6% to 81.6%** by building verifier-guided majority voting (3 candidates, 2-of-3 agreement) with adaptive retries that escalate the token budget and re-prompt with verifier feedback.
- Added **7 points** to the final score (61 to 68) by auditing 7,500 distilled examples, training LoRA SFT/DPO adapters, and deploying them only as a fallback on unresolved multiple-choice questions.

Persona Chatbot & Discord Bot Platform

Python, PyTorch, Hugging Face, llama.cpp, discord.js

- Built a 16,581-conversation training set from **330,000** Discord messages (8 years) by scrubbing personal info, merging alt accounts, and segmenting history.
- Fit **36 personas** into one 1.7B model served locally at 4-bit by fine-tuning Qwen3-1.7B with QLoRA on a single 12 GB GPU (64,000 examples), gated by a DPO pass that ships only if it beats SFT on held-out chats.
- Kept each persona's voice and memory consistent by building two-index retrieval (own lines for voice, past chats for facts) scoped to DMs it was in; the bot serves up to 5 users with no open ports.

Handwritten Digit Recognition

Python, NumPy, JavaScript

- Reached **95.4%** test accuracy, classifying 1,000 MNIST digits in ~ 0.25 s, by building a vectorized 1-NN classifier from scratch over 7,500 training images, benchmarked against scikit-learn's tree search.
- Made the model explorable in any browser by porting the search to JavaScript: draw a digit and see its five nearest training images.

Self-Hosted Game Server Infrastructure

Personal Homelab

- Migrated a live Minecraft network serving **30+** users (proxy + 3 game servers) into Docker in one day with **zero data loss** by writing a runbook and a backup-first cutover script with rollback.
- Freed **~7.5 GB** of a 16 GB host's RAM for a co-hosted LLM by starting the two largest worlds on demand and stopping them when empty; protected data with hourly, restore-tested restic backups to an append-only second machine.
- Exposed **zero** admin ports by moving the web panel behind Cloudflare Tunnel with 2FA and SSH/dashboards to Tailscale; closed a Docker firewall bypass and ended recurring SSD resets by tracing them to an NVMe APST firmware bug.

EDUCATION

University of California, San Diego

Expected Jun. 2027

B.S., Computer Engineering

La Jolla, CA

- **Relevant Coursework:** Machine Learning: Learning Algorithms; Deep Learning (CSE 151B); Probabilistic Reasoning in AI; Computer Design and Architecture (Verilog, C, MIPS); Data Networks; Data Structures (Java)

TECHNICAL SKILLS

- **Languages:** Python, Java; C, C++; JavaScript; SystemVerilog, Verilog; MIPS Assembly; MATLAB; Bash
- **ML & AI:** PyTorch; vLLM; Hugging Face (TRL, PEFT); LoRA/QLoRA fine-tuning (SFT, DPO); RAG; llama.cpp; Ollama; NumPy
- **Linux & DevOps:** Ubuntu Server; Docker; systemd; LVM; restic; Tailscale; Cloudflare Tunnel; ufw; fail2ban; iptables; tcpdump; SSH
- **Hardware & Tools:** ModelSim; Quartus; self-checking testbenches; Git; Maven; Paper API; Vim; LaTeX